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Expt.-9: Apply PCA and LDA to reduce dataset dimensions

Machine Learning Lab

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import time

from sklearn import datasets
from sklearn.preprocessing import StandardScaler
from sklearn.decomposition import PCA
from sklearn.discriminant_analysis import LinearDiscriminantAnalysis

from sklearn.model_selection import train_test_split
from sklearn.neighbors import KNeighborsClassifier
from sklearn.metrics import accuracy_score
```

```
iris = datasets.load_iris()
x=iris.data
y=iris.target

features_names=iris.feature_names
print(iris.feature_names)
print()
print(iris.target_names)
```

```
['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)', 'petal width (cm)']

['setosa' 'versicolor' 'virginica']
```

```
# Part I: Dataset Summary Table

print("DATASET SUMMARY")
print("Number of Samples:",x.shape[0])
print("Number of features:",x.shape[1])
print("Number of Classes:",len(np.unique(y)))
print(features_names)
print()
```

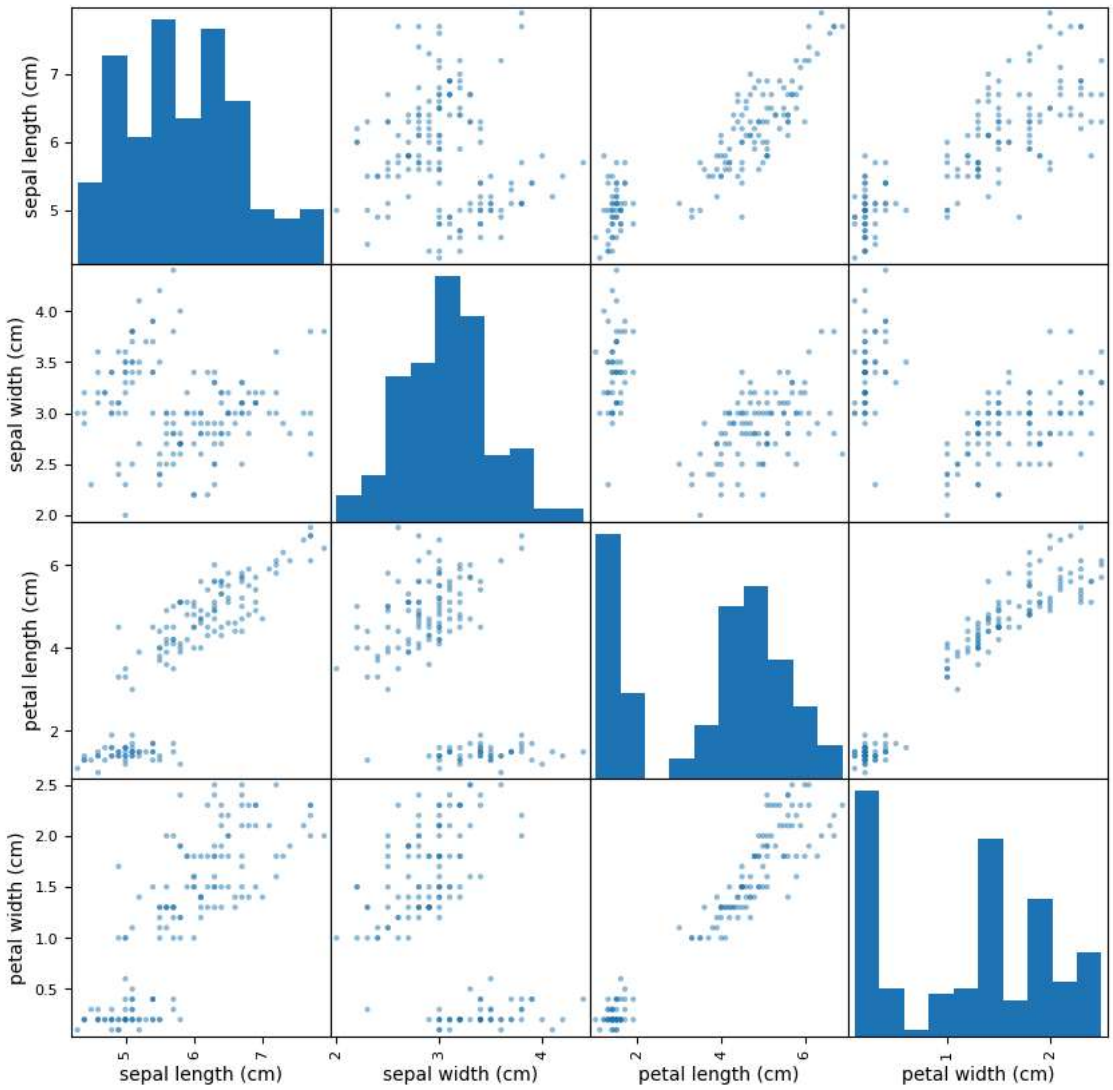
```
DATASET SUMMARY
Number of Samples: 150
Number of features: 4
Number of Classes: 3
['sepal length (cm)', 'sepal width (cm)', 'petal length (cm)', 'petal width (cm)']
```

```
# Part II - FEATURE STATICS TABLE (BEFORE PCA)
df=pd.DataFrame(x, columns=features_names)

print("FEATURE STATISTICS")
stat_table=pd.DataFrame({
    "Mean": df.mean(),
    "Standard Deviation": df.std()
})

print(stat_table)
print()
print(df.head())
print()
```

```
FEATURE STATISTICS
                      Mean  Standard Deviation
sepal length (cm)  5.843333            0.828066
sepal width (cm)   3.057333            0.435866
petal length (cm)  3.758000            1.765298
```

```
petal width (cm)   1.199333            0.762238

   sepal length (cm)  sepal width (cm)  petal length (cm)  petal width (cm)
0                5.1               3.5                1.4               0.2
1                4.9               3.0                1.4               0.2
2                4.7               3.2                1.3               0.2
3                4.6               3.1                1.5               0.2
4                5.0               3.6                1.4               0.2
```

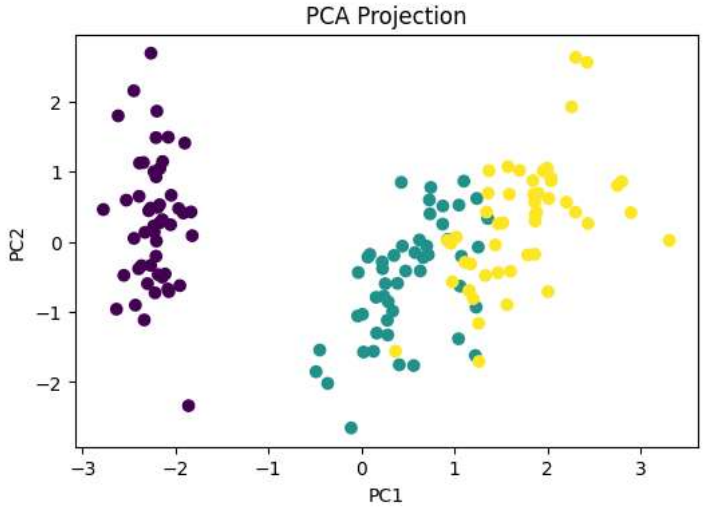
```
pd.plotting.scatter_matrix(
    df.iloc[:,0:4],
    figsize=(10,10),
    diagonal='hist'
)

plt.show()
```




```
# Part III - PCA

sc=StandardScaler()
x_scaled=sc.fit_transform(x)

pca=PCA(n_components=2)
x_pca=pca.fit_transform(x_scaled)
```

```
plt.figure(figsize=(6,4))
plt.scatter(x_pca[:,0],x_pca[:,1],c=y)
plt.title("PCA Projection")
plt.xlabel("PC1")
```

```
plt.ylabel("PC2")
plt.show()
```



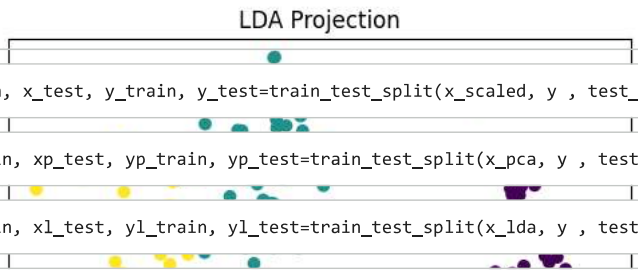

```
explained_variance=pca.explained_variance_
variance_ratio=pca.explained_variance_ratio_
cumulative_variance=np.cumsum(variance_ratio)
```

```
pca_table=pd.DataFrame({
    "Eigenvalue": explained_variance,
    "Explaind Variance (%)": variance_ratio*100,
    "Cumulative Variance (%)": cumulative_variance*100
})

print("PCA VARIANCE TABLE")
print(pca_table)
print()
```

```
PCA VARIANCE TABLE
   Eigenvalue  Explaind Variance (%)  Cumulative Variance (%)
0    2.938085              72.962445                72.962445
1    0.920165              22.850762                95.813207
```

```
# PART IV - LDA

lda=LinearDiscriminantAnalysis(n_components=2)
x_lda=lda.fit_transform(x_scaled, y)
```

```
plt.figure(figsize=(6,4))
plt.scatter(x_lda[:,0],x_lda[:,1],c=y)
plt.title("LDA Projection")
plt.xlabel("LD1")
plt.ylabel("LD2")
plt.show()
```




```
x_train, x_test, y_train, y_test=train_test_split(x_scaled, y , test_size=0.3, random_state=42)
```

```
xp_train, xp_test, yp_train, yp_test=train_test_split(x_pca, y , test_size=0.3, random_state=42)
```

```
xl_train, xl_test, yl_train, yl_test=train_test_split(x_lda, y , test_size=0.3, random_state=42)
```

```
model=KNeighborsClassifier()
#Original Data
start=time.time()
model.fit(x_train,y_train)
train_time_original=time.time()-start

y_pred=model.predict(x_test)
acc_original=accuracy_score(y_test, y_pred)
#PCA Data
start=time.time()
model.fit(xp_train,yp_train)
train_time_pca=time.time()-start

y_pred=model.predict(xp_test)
acc_pca=accuracy_score(yp_test, y_pred)
#LDA Data
start=time.time()
model.fit(xl_train,yl_train)
train_time_lda=time.time()-start

y_pred=model.predict(xl_test)
acc_lda=accuracy_score(yl_test, y_pred)
```

```
accuracy_table=pd.DataFrame({
    "Dataset Version":["Original", "PCA Reduced","LDA Reduced"],
    "Accuracy": [acc_original, acc_pca, acc_lda],
    "Training Time (seconds)":[train_time_original, train_time_pca, train_time_lda]
})
print("MODEL PERFORMANCE TABLE")
print(accuracy_table)
```

```
MODEL PERFORMANCE TABLE
  Dataset Version  Accuracy  Training Time (seconds)
0        Original  1.000000                 0.003081
1     PCA Reduced  0.955556                 0.001469
2     LDA Reduced  1.000000                 0.001282
```

Start coding or generate with AI.